

# What Is an Operating System?

A list of commonly used operating systems includes Microsoft Windows, MacOS, UNIX, Linux, and many others. They share many common features, and, at the same time, exhibit many differing aspects.

What is an operating system? The [Merriam-Webster dictionary](#) defines an operating system as:

*"software that controls the operation of a computer and directs the processing of programs."*

Although that is a concise definition, it really does not say much; let's expand on the definition.

The heart of an operating system is its kernel. This is sometimes a very big program, but still a program. In fact, the word Linux technically only refers to the kernel, and not the whole operating system.

The kernel has privileged features such as the ability to directly control devices connected to it. Devices such as a mouse, keyboard, screen, memory, storage, and many, many more are controlled by the kernel.

If the outside world would like to access these devices controlled by the Linux kernel, there must be an interface to interact with it; this interface is in fact another program, a program that can communicate with the kernel to cause some action in the hardware to occur (e.g., actions like light a bulb, read a key button being depressed, etc). Assembling the components so far, we have a kernel that controls the hardware and a program that wants to access the hardware, but the program is unaware of how to communicate with the devices. We need an interface between the kernel and the user program.

The system call is the fundamental interface between an application and the Linux kernel.

This interface between the user program and the Linux kernel is the system call (or syscall), which is privileged code that can make requests to the Linux kernel. The kernel accepts the properly formatted request, performs the requested action and issues a return code indicating access or failure.

An example is a user program that would like to change to another directory (folder): the program calls the function `chdir()` which invokes the `syscall chdir()` which is in the Linux kernel. A success or failure code (return code) is provided to the user program by the Kernel and to the function for examination.

It is commonplace to refer to Linux as not only the kernel, but as incorporating a complete environment. There are many different Linux distributions that are designed for particular purposes.